

6	(a)	(i)	Since the <u>velocity is always right angle to the magnetic field</u> , by Fleming's left hand rule, <u>the magnetic force always be right angle to its velocity</u> . The force <u>only changes the direction of the motion and not its speed</u> . This results in circular motion.	B1 B1
		(ii)	Out of page	B1
		(iii)	The magnetic force provides the centripetal force $Bqv = \frac{mv^2}{r}$ $B = \frac{mv}{qr} = \frac{(20 \times 1.66 \times 10^{-27})(5.6 \times 10^5)}{(1.6 \times 10^{-19})(0.051)}$ $B = 2.3 \text{ T}$	B1 C1 A1
	(b)	(i)	Upwards	B1
		(ii)	$qE = Bqv$ $E = Bv = (2.3)(5.6 \times 10^5)$ $= 1.3 \times 10^6 \text{ V m}^{-1}$	C1 A1

7	(a)	(i)	$m_H + m_H = m_V + m_V$	
---	-----	-----	-------------------------	--